Troubleshooting Tools.

Task 1. Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

Answer:

1. Press Windows + R, type cmd and press Enter to open the Command Prompt.
2. Type the command below and press Enter.
3. In the result, find the adapter you are using (Wireless LAN adapter Wi-Fi or Ethernet adapter), then note the "IPv4 Address" and the "Default Gateway" lines.



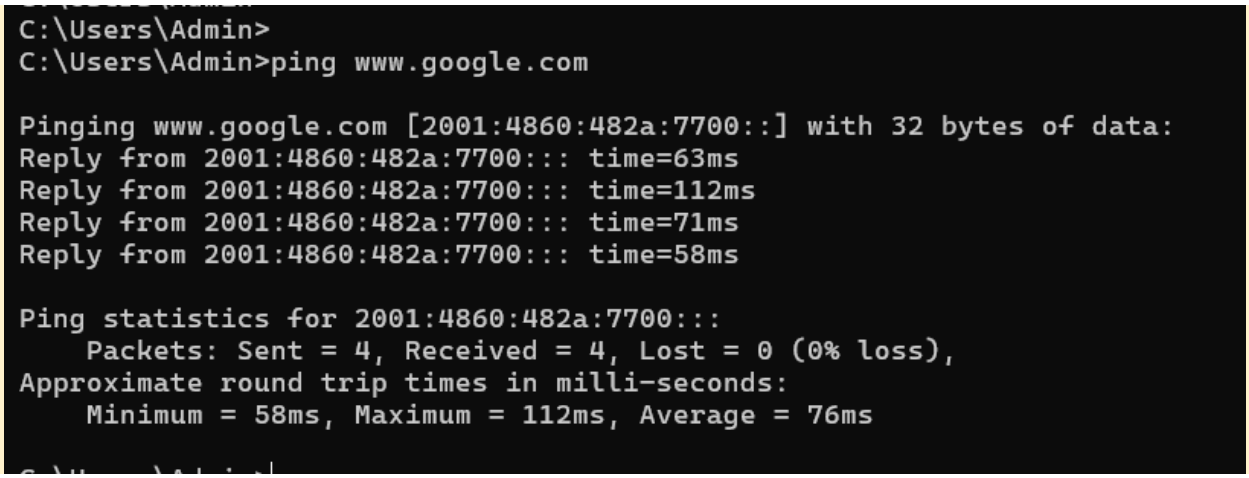
Task 2. Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

Answer:

`ping` www.google.com

Ping sends 4 test packets and shows the reply time of each. At the end, "Approximate round trip times in milli-seconds" shows Minimum, Maximum and Average. The Average value is the answer. "0% loss" means every packet got a reply, so the connection works.

My average response time:



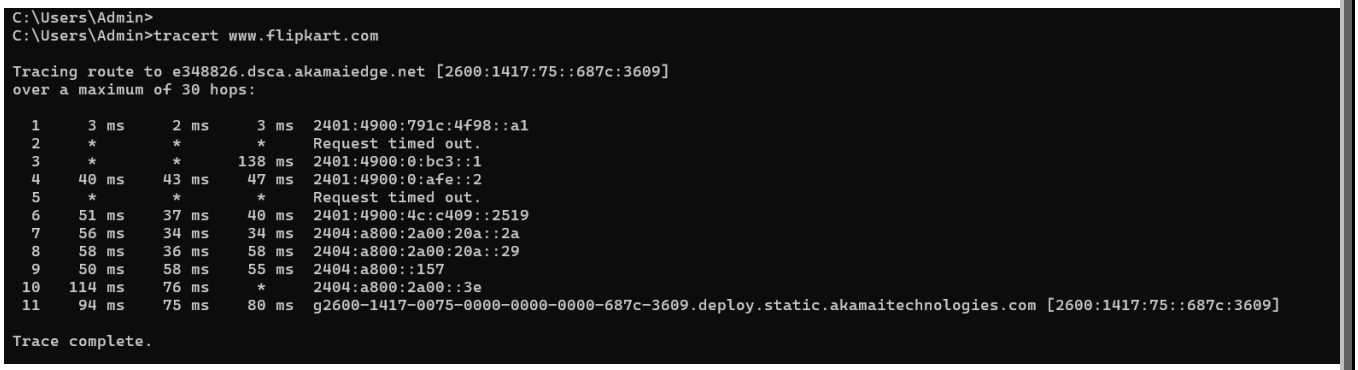
Task 3. Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

Answer:

`tracert` www.flipkart.com

Each numbered line in the result is one hop (one router on the path). The number on the last line, where the destination IP of Flipkart appears, is the total number of hops. Lines with "Request timed out" (* * *) still count as hops, because some routers do not reply to tracert. The default maximum is 30 hops. On Linux or Mac use: traceroute www.flipkart.com

Number of hops to reach www.flipkart.com



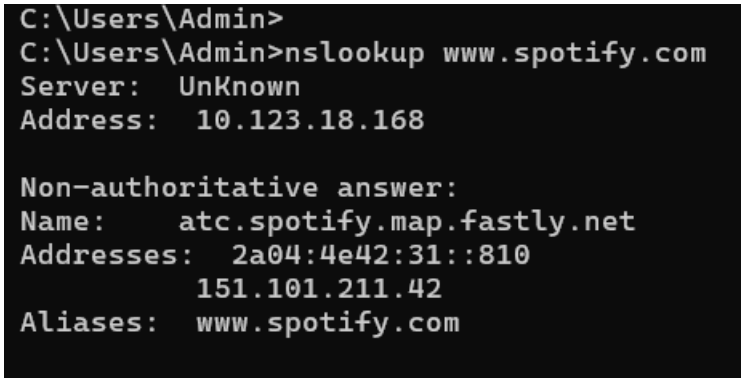
Task 4. Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

Answer:

`nslookup` www.spotify.com

In the result, the lines under "Name: www.spotify.com" (or its alias) show "Address" or "Addresses". Those are the IP addresses of Spotify. The first "Server / Address" lines show your DNS server. Big sites often show more than one IP, and the IP can change by location and time.

IP address of www.spotify.com:



What nslookup helps verify (one line): It checks whether DNS is working, by showing if a website name is correctly converted into an IP address and which DNS server answered.

Task 5. A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.

Answer:

Step	Command	What it helps me check
1	ipconfig	Whether the laptop has a valid IP address, default gateway and DNS server. If the IP starts with 169.254.x.x or the gateway is empty, the laptop is not connected properly to the network (a Wi-Fi or DHCP problem), and that is the cause.
2	ping 8.8.8.8 (a public IP, without a name)	Whether the laptop can reach the internet. If it replies, the network connection is working. If it shows "Request timed out", the problem is the network, the router or the ISP, and I can check the gateway with ping next.
3	nslookup www.zomato.com	Whether DNS can turn the name into an IP address. If ping 8.8.8.8 works but nslookup fails ("DNS request timed out" or "Non-existent domain"), then it is a DNS problem, and the fix is to change the DNS to 8.8.8.8 or flush it with ipconfig /flushdns.

If all three work fine, the problem is not the network or DNS. Then I would run tracert www.zomato.com to see where the path stops, and also check the browser, proxy, firewall or whether the website itself is down.